Test-1

Name : D. Hussaiah

Regno : 192425300

Course code : MLA0201

Course : fundamental of M.L

1) The Evidance. $p(+) = 0.90 \times 0.03 + 0.06 + 0.97$

$= 0.0270 + 0.582$

$= 0.0852$

$P(-) = 1 - 0.0852 = 0.9148$

posterlars $p(c|+) = 0.0270 / 0.0852$

$= 0.3169$

$p(c|+) = 0.6831$

$p(c|-) = 0.10 \times 0.03 / 0.9148$

$= 0.9967$

Conditional risks : $R(t|+) = 1.3662$

$R(no|t) = 50 \times 0.3169 = 15.48$

$R(t|-) = 1.9934$

$R(no|t) = 50 \times 0.0033 = 0.1640$

Minimum Risk Decision :

(positive → treat) (1.3662 < 15.485)

Bayes risk :

$P(t,-) = 2 \times 0.0582 + 50 \times 0.0030 = 0.2664$

(0-1 rule):

$P(C|x) > 0.5$

since $P(C|t) = 0.3169$

loss matrix $= 50 \times 0.03$

$= 1.5$

Second postive test:

$\Rightarrow 6.959 / 7.959$

$= 0.8744$

Evaluation:

The minimum risk rule reduce expected lost from 1.5 to 0.2664 a reduction of 82.24%

1)

X

2) i) Given 48, 52, 55, 47, 53, 51

Mean

$\bar{x} = \frac{300}{6} = 50$

ML Variance

$\Rightarrow p(x) = 0.32, 0.38, 0.30$

$p(y) = 0.68, 0.32$

$H(x) = -\sum p \log_2 p = 0.5260 + 0.5304 + 0.5211$

$= 1.5776$

$H(y) = 0.3784 + 0.5260$

$= 0.9044$

eg: $-0.32 \log_2 0.32 = 0.5260$

$H(x|y) = 0.5211 + 0.1129 + 0.5142 + 0.3322 + 0.4644$

$= 2.2770$ bits

$= 2.2701$

$$H(y|x) = H(x,y) - H(x)$$
$$= 2.2770 - 1.5776$$
$$= 0.6994 \text{ bits}$$

$$I(x;y) = H(y) - H(y|x)$$
$$= 0.2050 \text{ bits}$$

Cross entropy:

$$H(p-x, q) = -\sum p-x \log_2 q = 0.32\,(1.000)$$
$$+ 0.38\,(1.7370)$$
$$= 1.6766 \text{ bits}$$

KL divergence:

$$D_{KL}(p-x \| q) = 1.6766 - 1.5776$$
$$= 0.0990 \text{ bits}$$

$$P(y = \text{yes} \mid x = \text{High}) = 0.20/0.30$$
$$= 0.6667$$

Interpretation & conclusion:

The mutual information of 0.205 bits is positive.

(c) mean := $\frac{48+52+55+47+53+51}{6}$

$= 51$

ML variance:

$(-3)^2 + 1^2 + 4^2 + (-4)^2 + 2^2 + 0 = 46$ ML

variance $\Rightarrow \frac{46}{6} = 7.667$

un biased variance $s^2 \Rightarrow \frac{46}{5} = 9.20$

$s = 3.033$°C

$n = 6$

$P(x > 55)$: $z = (55-51) / 2.769$

$= 1.445$

$P = 1 - \Phi(1.445)$

$= 0.0743$

$P(48 < x < 54) = 1 - 0.1.8214$

$= 0.724144$

(ii) Mean $\bar{x} = 30$

$y = 51$

$S_{xx} = 1000$

P(spam) = 40/100

= 0.40

P(legit) = 60/100

= 0.60

Smooth likelihood spam:

spam = 42 , offer 29/42 = 0.6905 free = 25/42 = 0.59

meet 5/42 = 0.1190 , 0.1129, 0.0645,

Class scores:

X = (offer = 1, free = 1, meeting = 0)

Span = 0.40 X 0.6905 X 1 - 0.1190

= 0.1148

legit = 0.60 X 0.1129 X 0.00197

= 0.00197

P(spam|x) = 0.1448 / 0.1448 + 0.00197

= 0.9866

Cost-sensitive Decision:

P(spam|x) : 1 > P(legit|x) : 5 = 0.8333

since 0.9866 > 0.8333

classify as spam.

Test metrics:

Accuracy = 45 + 130 / 200

= 87.5%.

precision = 45 / (45 + 10)

= 81.82%.

Recall = 45 / (45 + 15)

= 75

F_1 = 2PR / (P+R)

= 78.26%.

=> cost = 5 × FP + 1 × FN

= 5 × 10 + 15

= 65

Interpretation & conclusion:

The new-mail is classified as spam with posterior 0.9866.

Weight update:

$$w_{22} = 0.5 - 0.8(1.0)$$
$$= 0.4980$$
$$b_2 = -0.1 - 0.0080 = -0.1080$$

Second forward:

$$z\text{-}o = 0.6281(0.5295) - 0.3661(0.6315) = 0.3548$$
$$y = 0.5878$$
$$E = 1/2(0.4122)^2 = 0.850$$

Evaluation:

The error falls from 0.0946 to 0.0850 a reduction of 10.2%.

